If significant congestion or delays is to occur on-site, which is unacceptable, works must cease and traffic control measures removed from the roadway or scaled back, where safe to do so. If any unsafe areas remain, additional protection must be implemented in accordance with the appropriate aftercare drawing. Once congestion eases, works can continue or alternatively, return to site for a later date and/or time.

4.1.6 End of Queue Treatment

The Traffic Management Assessor or Delivery Supervisor must assess end of queues and provide mitigating treatments as per the details provided in the Main Roads Code of Practice. This should be provided as supplementary information which accompanies the selected Traffic Guidance Schemes. The requirements and details of traffic queueing varies dependant on the project, location, and stage of the works.

Where required, the following should be used as a guide for determining queue lengths, further information is provided within AGTTM Part 3: Static Worksites Section 4.8 and Main Roads Code of Practice Section 6.8.2. The following information is required to determine the estimated queue lengths:

- The speed of traffic.
- The road environment (vertical / horizontal curves, road surface, etc.).
- The sight distance for road users to Traffic Control position and 'Prepare to Stop' signage.
- Driver reaction times (generally 2.5 seconds)
- The traffic volume (should consider peak traffic volumes at or near the location of works).
- The traffic composition (large vehicles require extra stopping and extend queue lengths).
- Work times and duration.
- Expected time traffic will be stopped.
- Worksite length

Once all of the above information has been gathered the expected queue length can be determined by using the following steps:

1. Determine the hourly traffic volume in the direction of travel at the time of the works (where volumes are not available, the assessor should attend site and count vehicles for a 5 minute period).
2. Divide the hourly traffic volume by 60 to determine the vehicle expected every minute.
3. Determine the length of time to the nearest minute that vehicles will be required to stop, this should include hold times for the work and travel time for opposing traffic direction to clear the road based on the length and speed of the worksite.
4. Multiply this number by the vehicles expected per minute.
5. Determine the types of vehicles that will be using the road and multiply its length by the number (including a 3m space between each vehicle).

Table 15: Approximate Vehicle Lengths MRWA Code of Practice Table 9

Vehicle Type	Approximate Length
Car	5.5m
Truck / Bus	19m
Trucks (RAV 2 to 4)*	27.5m
Road Train / B-Double (RAV 5 to 7)*	36.5m
Triple Road Train / Large Combination (RAV 9 & 10+)*	53.5m

**RAV classification assumes the worst-case length/combination. i.e 27.5m trucks may be used on RAV 2 to 7 however worst case from RAV 5 to 7 is 36.5m.*

Refer to the Main Roads Code of Practice for how to apply the calculated end of queue with the requirements for end of queue treatments. Where additional signage or devices are required for End of